

Performance Estimation Problems (PEPs)

Systematic analysis and design of first-order optimization algorithms
via the performance estimation framework

François Glineur, Adrien Taylor



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| Concrete example: analysis of a gradient step

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ (continuously differentiable). Find $x_\star \in \mathbb{R}^d$ such that

$$f(x_\star) = \min_{x \in \mathbb{R}^d} f(x),$$

with f is L -smooth and μ -strongly convex ($f \in \mathcal{F}_{\mu,L}$).

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Question: what *a priori* guarantees after N iterations?

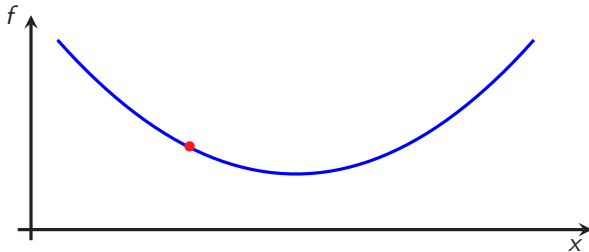
Examples: what about $f(x_N) - f(x_\star)$, $\|\nabla f(x_N)\|$, $\|x_N - x_\star\|$?

| About the assumptions

A differentiable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is μ -strongly convex and L -smooth iff $\forall x, y \in \mathbb{R}^d$:

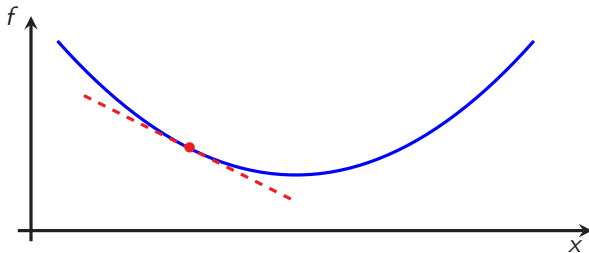
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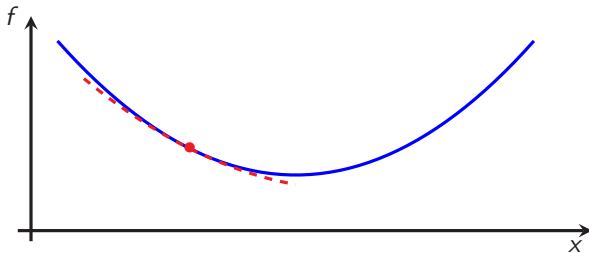
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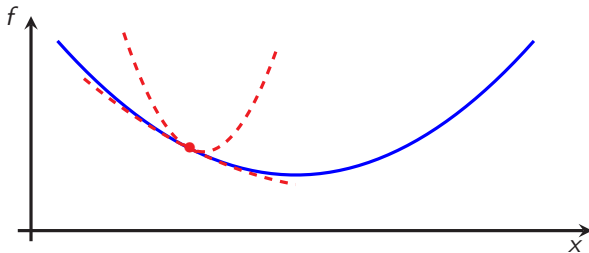


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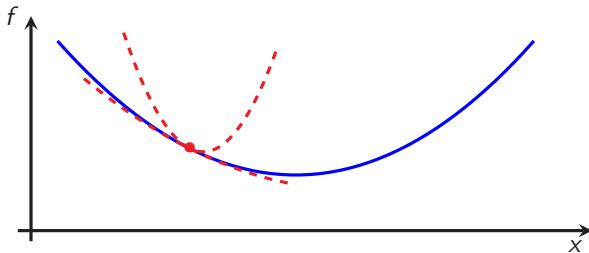
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(1&2) $\langle \nabla f(x) - \nabla f(y); x - y \rangle \geq \frac{1}{L+\mu} \|\nabla f(x) - \nabla f(y)\|^2 + \frac{\mu L}{L+\mu} \|x - y\|^2$.

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Toy example: What is the smallest τ such that:

$$\|x_1 - x_\star\|^2 \leq \tau \|x_0 - x_\star\|^2,$$

for all

- ◇ $d \in \mathbb{N}$, L -smooth and μ -strongly convex function f (notation $f \in \mathcal{F}_{\mu,L}$),
- ◇ x_0 , and x_1 generated by gradient step $x_1 = x_0 - \alpha \nabla f(x_0)$,
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$$\tau = \max_{f, x_0, x_1, x_\star} \frac{\|x_1 - x_\star\|^2}{\|x_0 - x_\star\|^2}$$

s.t. $f \in \mathcal{F}_{\mu,L}$

Functional class

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Functional class

Variables: $f, x_0, x_1, x_\star$.

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Algorithm

Parameters: μ, L, α .

$$\nabla f(x_\star) = 0$$

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- ◇ replacing them by

$$\begin{aligned} f_\star &\geq f_0 + \langle g_0, x_\star - x_0 \rangle + \frac{1}{2L} \|g_\star - g_0\|^2 + \frac{\mu}{2(1-\mu/L)} \|x_\star - x_0 - \frac{1}{L}(g_\star - g_0)\|^2 \\ f_0 &\geq f_\star + \langle g_\star, x_0 - x_\star \rangle + \frac{1}{2L} \|g_0 - g_\star\|^2 + \frac{\mu}{2(1-\mu/L)} \|x_0 - x_\star - \frac{1}{L}(g_0 - g_\star)\|^2. \end{aligned}$$

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- ◇ Same optimal value (no relaxation): **non-convex quadratic** problem.

| Semidefinite lifting

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◇ previous problem can be relaxed to 2×2 SDP

$$\begin{aligned} \max_{G, F} \quad & G_{1,1} + \alpha^2 G_{2,2} - 2\alpha G_{1,2} \\ \text{subject to} \quad & F + \frac{L\mu}{2(L-\mu)} G_{1,1} + \frac{1}{2(L-\mu)} G_{2,2} - \frac{L}{L-\mu} G_{1,2} \leq 0 \\ & -F + \frac{L\mu}{2(L-\mu)} G_{1,1} + \frac{1}{2(L-\mu)} G_{2,2} - \frac{\mu}{L-\mu} G_{1,2} \leq 0 \\ & G_{1,1} = 1 \\ & G \succcurlyeq 0 \end{aligned}$$

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- ◇ Assuming $x_0, x_*, g_0 \in \mathbb{R}^d$ with $d \geq 2$, same optimal value as original problem!
- ◇ For $d = 1$ same as original problem by adding $\text{rank}(G) \leq 1$.

| Dual problem

- ◇ Primal problem (abstract form)

$$\begin{aligned} \max_{G, F} \quad & \langle b_{\text{obj}}, F \rangle + \text{Tr}(A_{\text{obj}} G) \\ \text{subject to} \quad & \langle b_1, F \rangle + \text{Tr}(A_1 G) \leq 0 \\ & \langle b_2, F \rangle + \text{Tr}(A_2 G) \leq 0 \\ & \langle b_{\text{init}}, F \rangle + \text{Tr}(A_{\text{init}} G) = 1 \\ & G \succcurlyeq 0 \end{aligned}$$

- ◇ Lagrangian dual (provides upper bounds for any $\lambda_1, \lambda_2 \geq 0, S \succcurlyeq 0, \tau \in \mathbb{R}$):

$$\begin{aligned} \max_{G, F} \quad & \langle [b_{\text{obj}} - \lambda_1 b_1 - \lambda_2 b_2 - \tau b_{\text{init}}], F \rangle \\ & + \text{Tr}([A_{\text{obj}} - \lambda_1 A_1 - \lambda_2 A_2 - \tau A_{\text{init}} + S] G)) + \tau \end{aligned}$$

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- ◇ Direct consequence: for any $\tau \geq 0$ we have

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- ◇ Direct consequence: for any $\tau \geq 0$ we have

$$\|x_1 - x_\star\|^2 \leq \tau \|x_0 - x_\star\|^2 \text{ for all } f \in \mathcal{F}_{\mu, L}, \text{ all } x_0 \in \mathbb{R}^d, \text{ all } d \in \mathbb{N}, \text{ with } x_1 = x_0 - \alpha \nabla f(x_0).$$



$$\exists \lambda, \tau \geq 0 : \begin{bmatrix} \tau - 1 + \frac{\lambda L \mu}{L - \mu} & \alpha - \frac{\lambda (L + \mu)}{2(L - \mu)} \\ \alpha - \frac{\lambda (L + \mu)}{2(L - \mu)} & \frac{\lambda}{L - \mu} - \alpha^2 \end{bmatrix} \succcurlyeq 0$$

- ◇ Strong duality holds ($\exists$ Slater point): any valid worst-case convergence rate $\equiv$ valid dual feasible point ($\Downarrow$).

Dual problem

- ◇ Dual problem is

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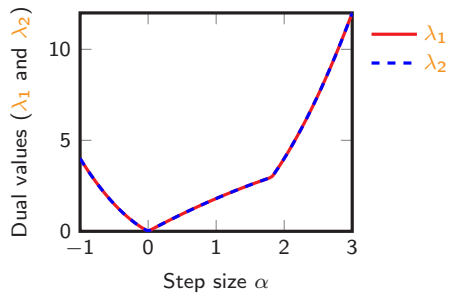
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Fix $L = 1$, $\mu = .1$ and solve the dual SDP for a few values of α .

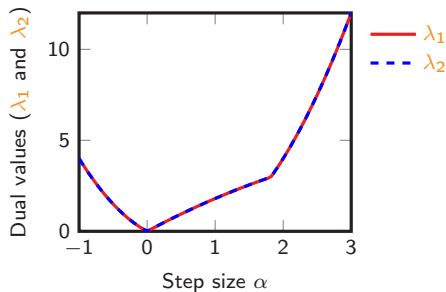
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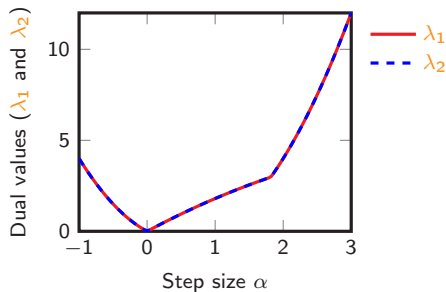
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Tutorial material:

<https://github.com/PerformanceEstimation/Tutorial-SIAM-OP26>

| Dual problem, sums-of-squares, and standard proofs

- ◇ Primal problem (abstract form)

$$\begin{aligned} \max_{G, F} \quad & \langle b_{\text{obj}}, F \rangle + \text{Tr}(A_{\text{obj}} G) \\ \text{subject to} \quad & \langle b_1, F \rangle + \text{Tr}(A_1 G) \leq 0 \\ & \langle b_2, F \rangle + \text{Tr}(A_2 G) \leq 0 \\ & \langle b_{\text{init}}, F \rangle + \text{Tr}(A_{\text{init}} G) = 1 \\ & G \succcurlyeq 0 \end{aligned}$$

- ◇ Dual problem (minimize upper bound)

$$\begin{aligned} \min_{\substack{\lambda_1, \lambda_2 \geq 0 \\ S \succcurlyeq 0 \\ \tau \in \mathbb{R}}} \quad & \tau \\ \text{subject to} \quad & b_{\text{obj}} - \lambda_1 b_1 - \lambda_2 b_2 - \tau b_{\text{init}} = 0 \\ & \lambda_1 A_1 + \lambda_2 A_2 + \tau A_{\text{init}} - A_{\text{obj}} = S \end{aligned}$$

- ◇ Remark that if $(\lambda_1, \lambda_2, \tau, S)$ is dual feasible, then by construction for all primal feasible (G, F) :

$$\begin{aligned} 0 & \geq \lambda_1 (\langle b_1, F \rangle + \text{Tr}(A_1 G)) + \lambda_2 (\langle b_2, F \rangle + \text{Tr}(A_2 G)) \\ & = \langle b_{\text{obj}}, F \rangle + \text{Tr}(A_{\text{obj}} G) - \tau (\langle b_{\text{init}}, F \rangle + \text{Tr}(A_{\text{init}} G)) + \text{Tr}(SG) \end{aligned}$$

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| Gradient descent... again!

Toy example: What is the smallest τ such that:

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Wooclap time!

<https://app.wooclap.com/THIKNZU?from=instruction-slide>



| Insight into proof structures from PEPs

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ (continuously differentiable) with $f \in \mathcal{F}$. Find $x_\star \in \mathbb{R}^d$ such that

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Then, proofs of the fact “Perf $\leq \tau$ Init” can be framed as: $\exists (\lambda_i)_i, \lambda_i \geq 0$, and $S \succcurlyeq 0$

$$0 \geq \sum_i \lambda_i \left(\text{Tr}(G A_i) + b_i^T F \right) = \text{Perf} - \tau \text{Init} + \text{Tr}(G S).$$